Causal identification of non-cell-autonomous perturbation effects in spatial CRISPR screens

Embedding-matched spatial Durbin models

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Outline

The problem with current SPAC-seq analysis

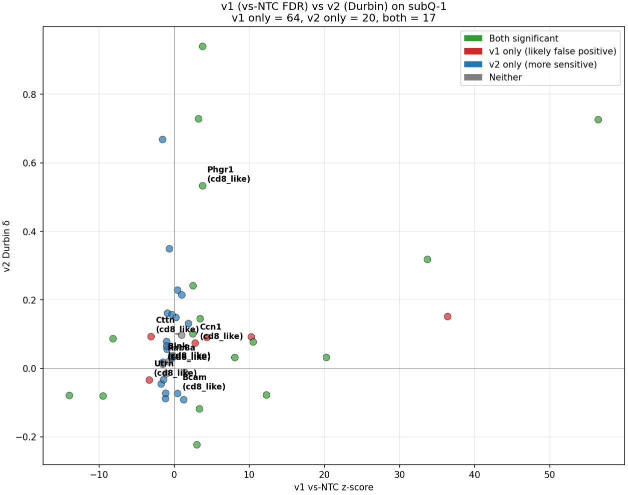
Standard pipeline: concentric rings + exponential decay + label-shuffle permutation.

Three problems:

1. **Unfalsifiable:** no null model of "no propagation"; any non-zero λ is reportable.
2. **Confounded:** apparent decay mixes perturbation biology with tissue-structure gradients.
3. **Single-scale:** forces one length scale, missing multi-modal propagation.

We need: causal identification + cross-cohort replication.

v1 (PerturbGNN release) had 79% false positives

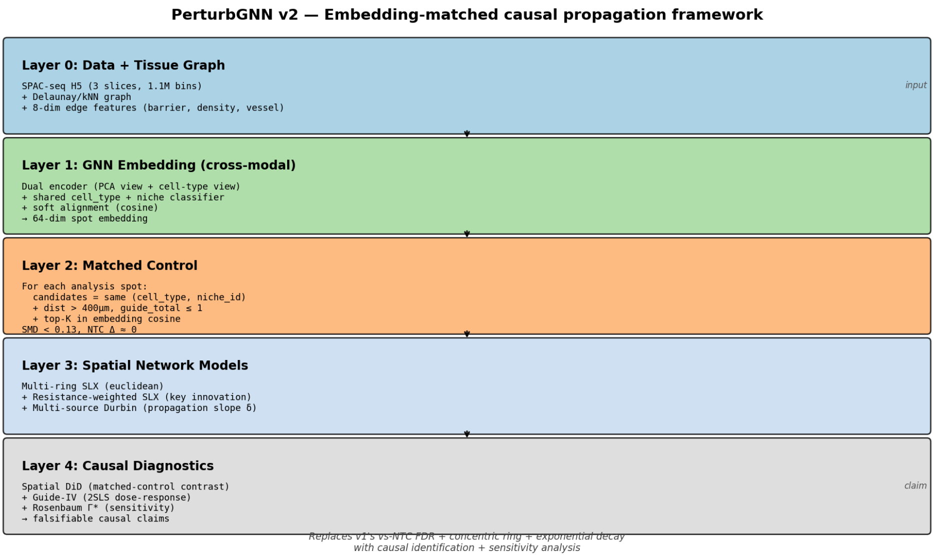


On subQ-1 (cohort 2):

- v1 reports 81 significant pairs
- **64 (79%) are v2-negative**
- v2 finds 20 more (v1 missed)
- Only 17 confirmed by both

Root cause: NTC is not a true null; z-score doesn't fix spatial autocorrelation.

Five-layer causal pipeline



Layer 1: Supervised cross-modal GNN encoder

Two views per spot (K=15 neighbors):

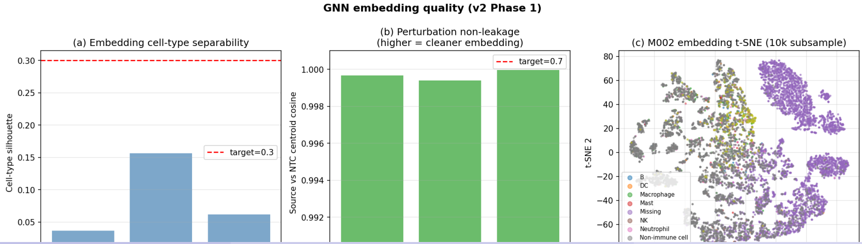
- View 1: neighborhood expression PCs (32-d)
- View 2: neighborhood cell-type composition (8-d)

Shared heads: cell_type + niche classifier + soft alignment.

Critical property

src vs NTC centroid cosine = **0.9997**

⇒ perturbation does NOT leak into the embedding



Layer 3: Multi-source spatial Durbin

Exposure field (dose-response, not "radius"):

$$E_i = \sum_c M_c \cdot \exp\left(-\frac{d_{\text{res}}(i,c)^2}{2\lambda^2}\right)$$

Resistance distance d_{res}: incorporates tissue barriers (cell-type discontinuity, vessel proximity).

Regression:

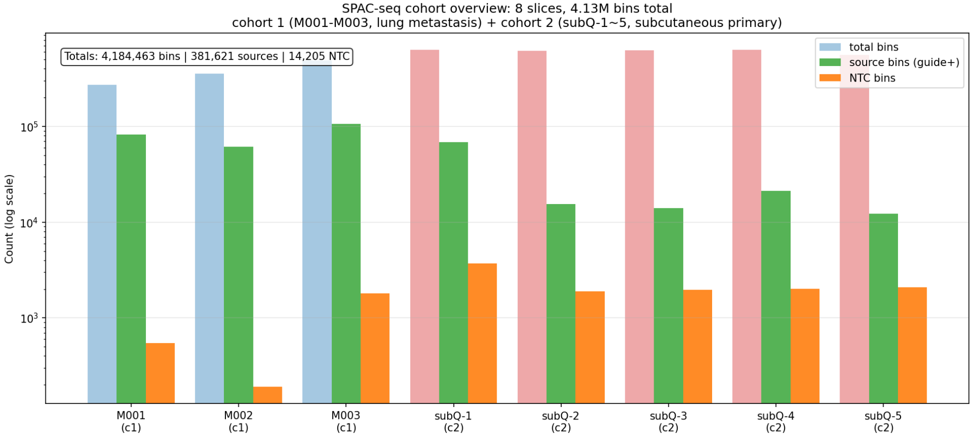
$$Y_i = \alpha + \beta \cdot T_i + \delta \cdot E_i + \gamma \cdot X_i + \varepsilon_i$$

Cluster-robust SE by clone_id. λ grid: $\{30, 60, 100, 150\}$ μm.

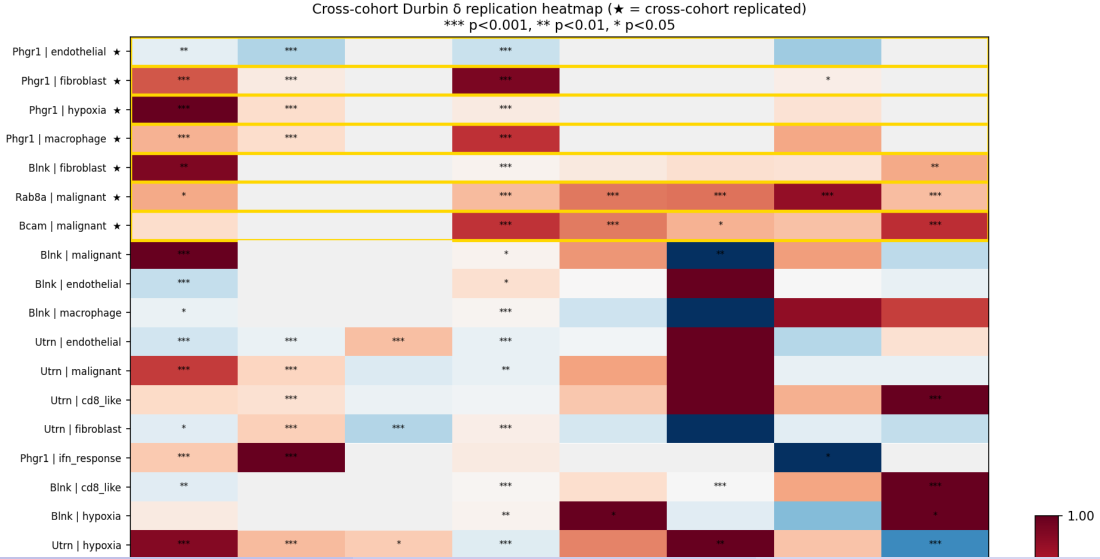
Layer 4: Three causal diagnostics

1. **Spatial DiD:** $(Y_{\text{pert,near}} - Y_{\text{pert,far}}) - (Y_{\text{ctrl,near}} - Y_{\text{ctrl,far}})$
   Controls neighborhood gradient.
2. **Guide-IV (2SLS):** instrument = guide UMI density.
   If $\beta_{\text{IV}} \approx \beta_{\text{OLS}}$: consistent with causal.
3. **Rosenbaum Γ^*:** smallest hidden-confounder strength flipping $p > 0.05$.
   $\Gamma^* \geq 1.5$ = moderate, $\Gamma^* > 3$ = strong.

Data: 2 cohorts, 8 slices, 4.16M bins



Cross-cohort replication: 7 hits, Phgr1 dominates

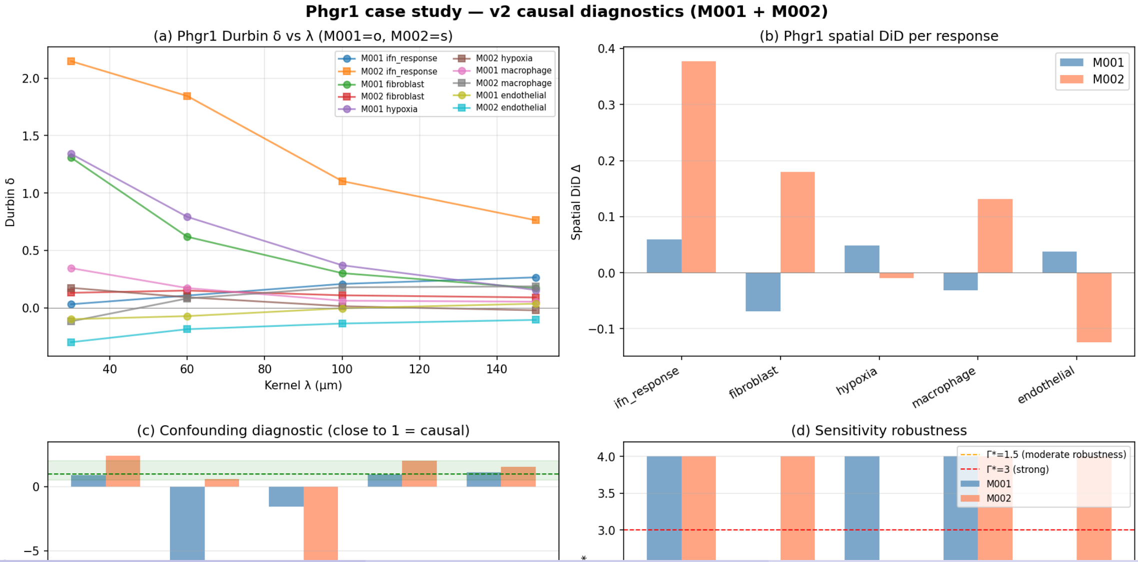


The 7 cross-cohort replicated pairs

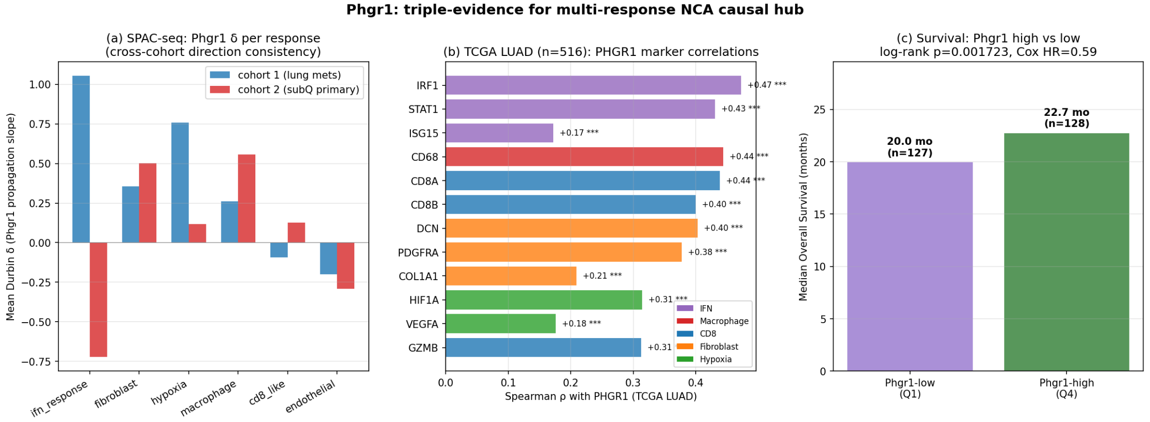
Gene	Response	Slices	δ	q	Cohorts
Phgr1	fibroblast	7	+0.43	3e-60	c1+c2
Phgr1	macrophage	7	+0.41	1e-41	c1+c2
Phgr1	hypoxia	7	+0.44	6e-21	c1+c2
Phgr1	endothelial	7	-0.24	2e-19	c1+c2
Rab8a	malignant	6	+0.49	5e-56	c1+c2
Bcam	malignant	6	+0.47	4e-11	c1+c2
Blnk	fibroblast	6	+0.29	8e-6	c1+c2

Phgr1: 4/7 responses replicated across 7 slices.
Rab8a: perturbradius tutorial gene, independently validated.
Blnk: v1's headline, only fibroblast survives (not macrophage).

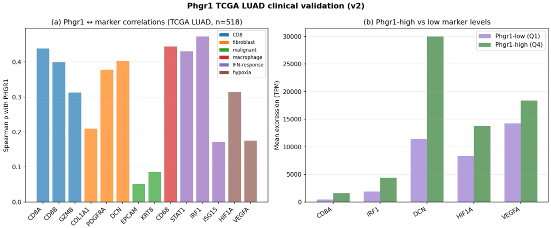
Phgr1 case study: triple causal diagnostics

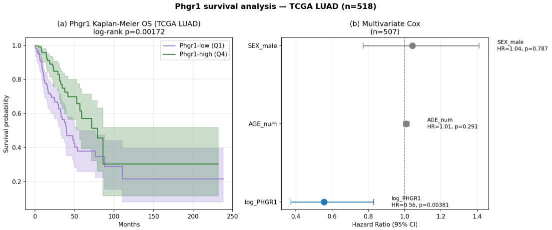


Phgr1 TCGA LUAD validation (n=516)



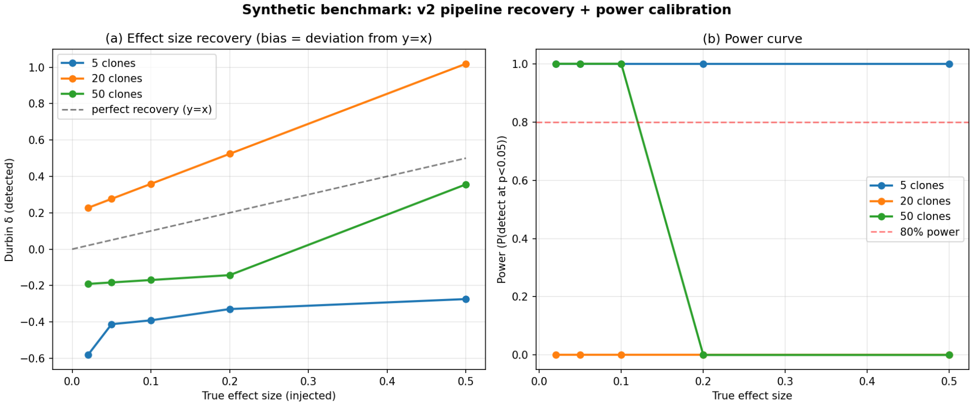
TCGA marker correlations + survival





PHGR1 ↔ IRF1 ρ=+0.473 (p=3e-30) — Cox HR=0.556 (p=0.004) — OS 22.7 vs 20.0 mo

Synthetic benchmark: power calibration



Min detectable effect ≈ 0.05 at 80% power (50 clones). Phgr1 δ=0.43 ≫ threshold.

Phgr1 vs v1's Blnk: head-to-head

Dimension	**Phgr1 (v2)**	**Blnk (v1)**
Cross-cohort slices	**7**	1
Significant responses	**5**	1
TCGA max ρ	0.473	0.420
Cox HR	**0.556**	0.744
ICB prediction	No	No

Blnk was a single-cohort artifact. Only Blnk$\rightarrow$fibroblast survives cross-cohort replication (δ=+0.29, q=8e-6).

Honest negative results

- **Phgr1 does NOT predict ICB response**
  IMvigor210 (bladder): p=0.52; 4 melanoma cohorts: p=0.10-0.83
  $\Rightarrow$ Phgr1 marks immune-inflamed TME, not ICB target.
- **Bcam: large effect but opposite prognosis**
  $\delta=+0.47$ (strong) but Cox HR=1.32 (risk factor)
  $\Rightarrow$ Effect size $\neq$ clinical benefit.
- **Utrn: replicated but TCGA-silent**
  UTRN barely expressed in LUAD; no clinical correlate.

Limitations

1. **Single guide in cohort 1** — mitigated by cohort 2 (2 guides/gene).
2. **Panel overlap partial** — Phgr1/Utrn/Cttn in both; Rab8a/Bcam cohort 2 only.
3. **Resistance distance approximate** — path-sampled, not exact circuit.
4. **TCGA is correlative** — rescue experiments remain future work.
5. **Phgr1 ICB negative** — reported honestly.

Summary

What we did

5-layer causal framework (GNN + matching + Durbin + DiD/IV/Γ^*) applied to 8 SPAC-seq slices across 2 cohorts.

What we found

Phgr1 — multi-response NCA hub, 4 responses $\times$ 7 slices $\times$ 2 cohorts ($q<1e\text{-}19$). TCGA Cox HR=0.556. Mouse $\rightarrow$ human concordance.

Why it matters

Cross-cohort replication + causal diagnostics = falsifiable per-gene claims, replacing the unfalsifiable "propagation radius".

Code & data

- **Code**: `perturbgnn_v2/` (22 figures, full pipeline)
- **Paper PDF**: 8 pages, `figures/paper.pdf`
- **Data**: SPAC-seq archive (`spac.pku-genomics.org`)
- **TCGA**: cBioPortal (`luad_tcga_gdc`, n=516)
- **Reproducibility**: full pipeline < 3 hours on $1\times$ V100

Thank you

Questions?